

CHI-SQUARE TESTS

new 3.14–3.15 • old 8.4–8.6 • ★ old 8.1

ZONE A · COUNTS, A RIGHT TAIL, AND A TEST SELECTOR

$$\chi^2 = \sum_{df=(r-1)(c-1)} \frac{(O - E)^2}{E}$$

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

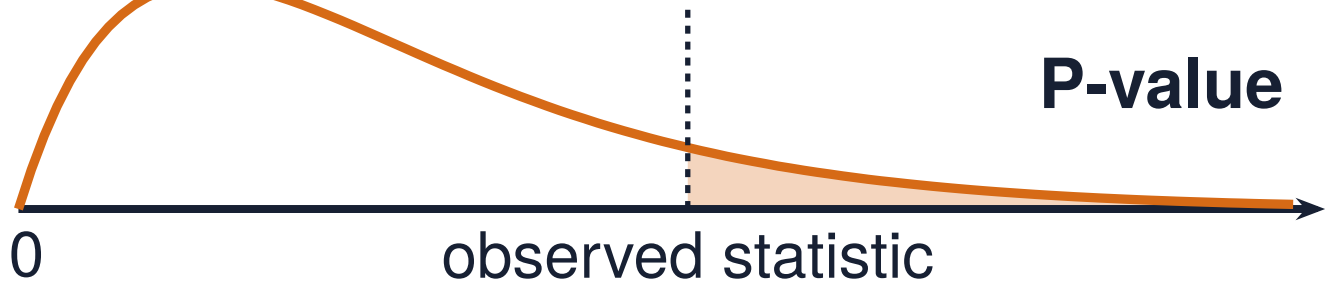
r : number of rows; c : number of columns.
Sum every cell’s contribution.

CHOOSE FROM THE DATA COLLECTION DESIGN

Test	Data shape	H_0 says
Homogeneity	One categorical variable; several populations or treatments	Same distribution in every population or treatment.
Independence	Two categorical variables; one population	The variables are not associated.
★ Goodness of fit	One categorical variable; claimed proportions	Matches the claimed distribution.

H_a : distributions differ; variables are associated; at least one claimed proportion differs, respectively.

SHADE TO THE RIGHT



Large values mean a larger mismatch. All chi-square test P-values are right-tail areas.

ZONE B · CONDITIONS, SYMBOLS, AND OUTPUT

1 · **Random** sample or assignment → 2 · **10%** $n < 0.10N$ without replacement
→ 3 · **Large Counts: every expected count** $E \geq 5$. Check each sampled population.

OBSERVED VS EXPECTED

O : observed count. E : count expected under H_0 .

$$\text{standardized residual} = \frac{O - E}{\sqrt{E}}$$

two-way test $df=(r-1)(c-1)$

Positive: excess; negative: shortfall.

COMPUTER-OUTPUT DECODER

$\chi^2 = 9.21, \quad df = 2, \quad P = .010$
9.21: total standardized discrepancy.
2: selects the null chi-square curve.
.010: under H_0 , chance of a statistic this large or larger.

ZONE C · SAY IT IN WORDS

“Because the P-value of ____ is [less/greater] than $\alpha =$ ____, we [reject/fail to reject] H_0 . There [is/is not] convincing evidence of an association between _____ and _____ for _____.”

For homogeneity or GOF, replace “association” with the matching H_a in context.

ZONE D · WATCH OUT + BEYOND

Check expected counts against 5.
Test: overall evidence. Residual: which cell.
Large |residual| marks a stronger departure; its square is the cell’s contribution.

★ **BEYOND THE EXAM**
GOF: $E = np_0$ for each category
 $df = k - 1$; k = number of categories.